

WASP-52b

R-1579

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_141030 · created 2026-08-01T23:44:20+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456960.7858 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.214 +/- 0.064
Transit depth (Rp/Rs)^2	4.57 +/- 2.72 [%]
Orbital Inclination (inc)	89.97 +/- 2.79
Airmass coefficient 1 (a1)	0.22 +/- 0.14
Airmass coefficient 2 (a2)	0.76 +/- 0.25
Scatter in the residuals of the lightcurve fit is	64.84 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	4.75
Transit Duration (day)	0.0871 +/- 0.0086

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.214 ± 0.064 vs 0.1646 (deviation 0.75σ)
Duration fit vs published	0.0871 d vs 0.0758 d (deviation 1.27σ)
Mid-time O–C	-0.1 min
Depth SNR	3.2
Residual scatter	64.84 %

Light curve

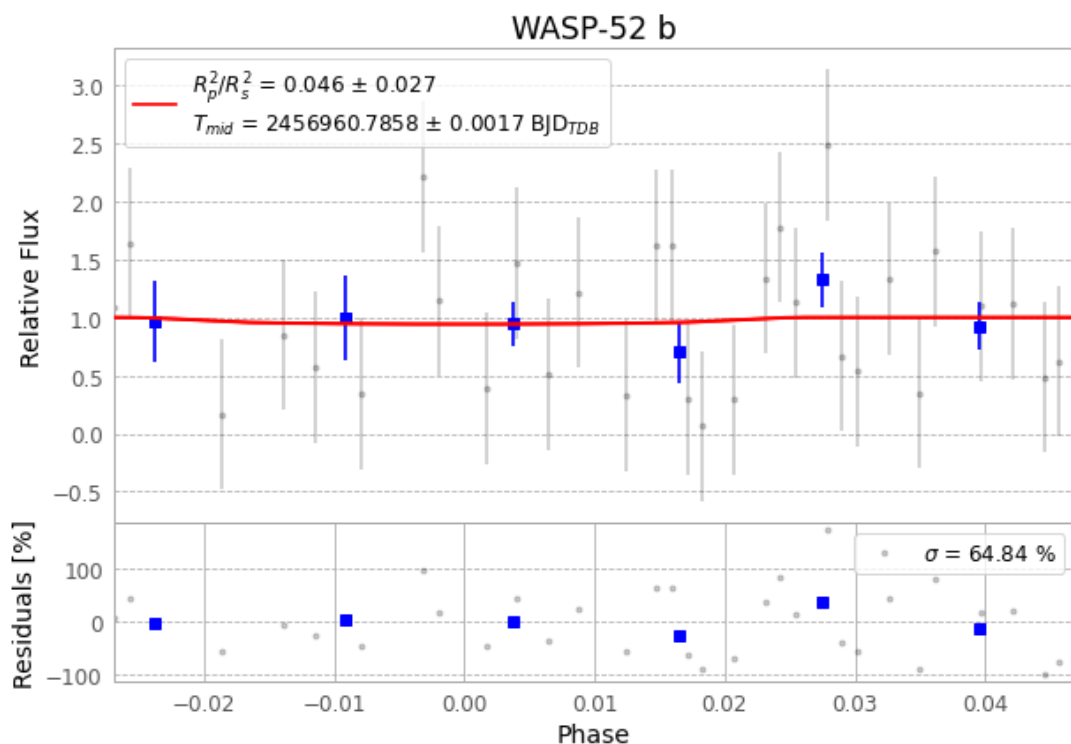


Figure 1 — FinalLightCurve_WASP-52 b_2014-10-29.png (SHA-256 9a5f5f39526d5d23...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [583, 426], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cd96f3ac249d3e5a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), WASP-52141030051212.FITS → WASP-52141030084512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-141030.log (ad2d64fa05bf...), FinalLightCurve_WASP-52 b_2014-10-29.png (9a5f5f39526d...), FinalLightCurve_WASP-52 b_2014-10-29.csv (9eb6732e8b02...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 23:44Z.